Advanced IR Project Part 2

Version 1.1

In the Project Part 2 assessment item of Advanced IR you will build a retrieval pipeline for web documents.

Grading

This project is assessed using the following criteria (20pts in total):

Implementation (10pts)

- **Code quality and comments (2.5pts)**
- **Comparison to a BM25 baseline and your own model from Part 1 (2.5pts)**
- **Solution thoughtfully designed to tackle the specific search problem (consideration of query and document properties, choice of model architectures, etc.) (5pts)**

Discussion (10pts)

- **Presentation, spelling, and grammar (5pts)**
- **Appropriate and thoughtful responses to prompts (5pts)**

Bonus

The top three submissions (not systems) to the leaderboard (see below) will be awarded bonus points:

- **1st place (5pts)**
- **2nd place (2.5pts)**
- **3rd place (2.5pts)**

Submission

Export the notebook as a PDF document with all of the code cells executed to show the outputs and the discussion filled out. Submit the PDF of this notebook to ILIAS (project link) by the 13.01.2026 23:55.

If you have developed any code outside of the notebook for the experiments, please include this additional code in a ZIP file alongside the original and PDF versions of this notebook.

Leaderboard

Project Part 2 also has a leaderboard component to see how your search system compares to everyone else in the class. You can find the leaderboard here: https://ir-leaderboard.scells.me/

To add a run to the leaderboard, visit https://ir-leaderboard.scells.me/upload?secret=YOUR_MATRICULATION_NUMBER You might want to go back to the first tutorial which explains more about this run format. You may also run `trec_eval` locally to validate that your run will be accepted by the online leaderboard.

Your name (as per ILIAS) will appear as the submitter, and the `runid` column in your run file (the last column) will appear as the system name. You can submit multiple systems and they will appear as separate rows in the leaderboard. Some decent baselines are already provided to try and beat. If you would prefer to use a pseudonym, please contact me by email (harrisen.scells@uni-tuebingen.de) and I will make the change for you; but please know that submitter names and system names cannot be changed once added to the leaderboard.

Tips

The test collection used in this lab is only available from bwJupyter. You likely will not have enough free credits to run your experiments on Google Colab, and bwJupyter provides much better GPUs than Colab that will make your experiments much faster to run.

In this project, you should try to be imaginative and go beyond what has been shown in the tutorial. You are

expected to read the pyterrier documentation to learn how to do things (like understanding the operators, using neural rankers, document processing, and the many other features available). You are of course not limited to pyterrier if you know about other tools you would like to use, like, e.g., pyserini.

If you are interested in running LLMs, huggingface has good tutorials and the GWDG can provide you with free API keys.

Furthermore, this project involves messy documents from webpages. These documents are much longer than the ones you've dealt with so far and will require some sort of processing to be able to use neural ranking models.

In [1]:

```
# These libraries are required to be able to access the dataset in pyterrier.
!pip install ir-datasets pyterrier git+https://github.com/tira-io/tira@d914806859f487037e7193a0773895937f39e8de#subdirectory=python-client

# Use this cell to install your other dependencies.
!pip install torch lightning-ir sentence_transformers transformers pyterrier-alpha openai codecarbon
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Collecting git+https://github.com/tira-io/tira@d914806859f487037e7193a0773895937f39e8de#subdirectory=python-client
  Cloning https://github.com/tira-io/tira (to revision d914806859f487037e7193a0773895937f39e8de) to /tmp/pip-req-build-r86pcm9h
  Running command git clone --filter=blob:none --quiet https://github.com/tira-io/tira /tmp/pip-req-build-r86pcm9h
  Running command git rev-parse -q --verify 'sha^d914806859f487037e7193a0773895937f39e8de'
  Running command git fetch -q https://github.com/tira-io/tira d914806859f487037e7193a0773895937f39e8de
  Running command git checkout -q d914806859f487037e7193a0773895937f39e8de
  Resolved https://github.com/tira-io/tira to commit d914806859f487037e7193a0773895937f39e8de
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Requirement already satisfied: ir-datasets in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (0.5.11)
Requirement already satisfied: pyterrier in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (1.0.1)
Requirement already satisfied: requests==2.*,>=2.26 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from tira==0.0.186) (2.32.5)
Requirement already satisfied: docker==7.*,>=7.1.0 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from tira==0.0.186) (7.1.0)
Requirement already satisfied: pandas in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from tira==0.0.186) (2.3.3)
Requirement already satisfied: packaging in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from tira==0.0.186) (25.0)
Requirement already satisfied: tqdm in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from tira==0.0.186) (4.67.1)
Requirement already satisfied: gitpython in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from tira==0.0.186) (3.1.46)
Requirement already satisfied: pyyaml in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from tira==0.0.186) (6.0.3)
Requirement already satisfied: pyterrier-alpha in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from tira==0.0.186) (0.16.4)
Requirement already satisfied: urllib3>=1.26.0 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from docker==7.*,>=7.1.0->tira==0.0.186) (2.6.2)
Requirement already satisfied: charset_normalizer<4,>=2 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from requests==2.*,>=2.26->tira==0.0.186) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from requests==2.*,>=2.26->tira==0.0.186) (3.11)
Requirement already satisfied: certifi>=2017.4.17 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from requests==2.*,>=2.26->tira==0.0.186) (2026.1.4)
Requirement already satisfied: beautifulsoup4>=4.4.1 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from ir-datasets) (4.14.3)
Requirement already satisfied: inscriptis>=2.2.0 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from ir-datasets) (2.7.0)
Requirement already satisfied: lxml>=4.5.2 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from ir-datasets) (5.4.0)
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Requirement already satisfied: numpy>=1.18.1 in /home/kpch/miniconda3/envs/py312/lib/pyth
on3.12/site-packages (from ir-datasets) (2.4.0)
Requirement already satisfied: trec-car-tools>=2.5.4 in /home/kpch/miniconda3/envs/py312/
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Requirement already satisfied: lz4>=3.1.10 in /home/kpch/miniconda3/envs/py312/lib/python
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Requirement already satisfied: warc3-wet>=0.2.3 in /home/kpch/miniconda3/envs/py312/lib/p
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Requirement already satisfied: zlib-state>=0.1.3 in /home/kpch/miniconda3/envs/py312/lib/
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Requirement already satisfied: pyarrow>=16.1.0 in /home/kpch/miniconda3/envs/py312/lib/py
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Requirement already satisfied: more_itertools in /home/kpch/miniconda3/envs/py312/lib/pyt
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Requirement already satisfied: deprecated in /home/kpch/miniconda3/envs/py312/lib/python3
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Requirement already satisfied: scipy in /home/kpch/miniconda3/envs/py312/lib/python3.12/s
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Requirement already satisfied: ir_measures>=0.4.1 in /home/kpch/miniconda3/envs/py312/lib
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Requirement already satisfied: pytrec_eval_terrier>=0.5.3 in /home/kpch/miniconda3/envs/p
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Requirement already satisfied: jinja2 in /home/kpch/miniconda3/envs/py312/lib/python3.12/
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Requirement already satisfied: statsmodels in /home/kpch/miniconda3/envs/py312/lib/python
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Requirement already satisfied: dill in /home/kpch/miniconda3/envs/py312/lib/python3.12/si
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Requirement already satisfied: soupsieve>=1.6.1 in /home/kpch/miniconda3/envs/py312/lib/p
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Requirement already satisfied: typing-extensions>=4.0.0 in /home/kpch/miniconda3/envs/py3
12/lib/python3.12/site-packages (from beautifulsoup4>=4.4.1->ir-datasets) (4.15.0)
Requirement already satisfied: cbor>=1.0.0 in /home/kpch/miniconda3/envs/py312/lib/python
3.12/site-packages (from trec-car-tools>=2.5.4->ir-datasets) (1.0.0)
Requirement already satisfied: wrapt<3,>=1.10 in /home/kpch/miniconda3/envs/py312/lib/pyt
hon3.12/site-packages (from deprecated->pyterrier) (2.0.1)
Requirement already satisfied: gitdb<5,>=4.0.1 in /home/kpch/miniconda3/envs/py312/lib/py
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Requirement already satisfied: smmap<6,>=3.0.1 in /home/kpch/miniconda3/envs/py312/lib/py
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Requirement already satisfied: MarkupSafe>=2.0 in /home/kpch/miniconda3/envs/py312/lib/py
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Requirement already satisfied: python-dateutil>=2.8.2 in /home/kpch/miniconda3/envs/py312
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Requirement already satisfied: pytz>=2020.1 in /home/kpch/miniconda3/envs/py312/lib/pytho
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Requirement already satisfied: tzdata>=2022.7 in /home/kpch/miniconda3/envs/py312/lib/pyt
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Requirement already satisfied: python-terrier>=0.13.0 in /home/kpch/miniconda3/envs/py312
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Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93 in /home/kpch/miniconda3/e
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Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90 in /home/kpch/miniconda3
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Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in /home/kpch/miniconda3/envs
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Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1 in /home/kpch/miniconda3/envs
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Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83 in /home/kpch/miniconda3/envs
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Requirement already satisfied: nvidia-curand-cu12==10.3.9.90 in /home/kpch/miniconda3/env
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Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90 in /home/kpch/miniconda3/e
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Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93 in /home/kpch/miniconda3/e
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Requirement already satisfied: nvidia-nccl-cu12==2.27.5 in /home/kpch/miniconda3/envs/py3
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Requirement already satisfied: nvidia-nvshmem-cu12==3.3.20 in /home/kpch/miniconda3/envs/
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Requirement already satisfied: nvidia-nvtx-cu12==12.8.90 in /home/kpch/miniconda3/envs/py
312/lib/python3.12/site-packages (from torch) (12.8.90)
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93 in /home/kpch/miniconda3/en
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Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3 in /home/kpch/miniconda3/envs
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Requirement already satisfied: triton==3.5.1 in /home/kpch/miniconda3/envs/py312/lib/pyth
on3.12/site-packages (from torch) (3.5.1)
Requirement already satisfied: datasets>=3.0.0 in /home/kpch/miniconda3/envs/py312/lib/py
thon3.12/site-packages (from lightning-ir) (4.4.2)
Requirement already satisfied: ir_datasets in /home/kpch/miniconda3/envs/py312/lib/python
3.12/site-packages (from lightning-ir) (0.5.11)
Requirement already satisfied: ir-measures in /home/kpch/miniconda3/envs/py312/lib/python
3.12/site-packages (from lightning-ir) (0.4.3)
Requirement already satisfied: lightning in /home/kpch/miniconda3/envs/py312/lib/python3.
12/site-packages (from lightning-ir) (2.6.0)
Requirement already satisfied: pandas in /home/kpch/miniconda3/envs/py312/lib/python3.12/
site-packages (from lightning-ir) (2.3.3)
Requirement already satisfied: jsonargparse>=4.27.7 in /home/kpch/miniconda3/envs/py312/l
ib/python3.12/site-packages (from jsonargparse[signatures]>=4.27.7->lightning-ir) (4.45.0
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Requirement already satisfied: sentencepiece in /home/kpch/miniconda3/envs/py312/lib/pyth
on3.12/site-packages (from lightning-ir) (0.2.1)
Requirement already satisfied: protobuf in /home/kpch/miniconda3/envs/py312/lib/python3.1
2/site-packages (from lightning-ir) (6.33.2)
Requirement already satisfied: tqdm in /home/kpch/miniconda3/envs/py312/lib/python3.12/si
te-packages (from sentence_transformers) (4.67.1)
Requirement already satisfied: scikit-learn in /home/kpch/miniconda3/envs/py312/lib/pytho
n3.12/site-packages (from sentence_transformers) (1.8.0)
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Requirement already satisfied: scipy in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from sentence_transformers) (1.16.3)
Requirement already satisfied: huggingface-hub>=0.20.0 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from sentence_transformers) (0.36.0)
Requirement already satisfied: numpy>=1.17 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from transformers) (2.4.0)
Requirement already satisfied: packaging>=20.0 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from transformers) (25.0)
Requirement already satisfied: pyyaml>=5.1 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from transformers) (2025.11.3)
Requirement already satisfied: requests in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from transformers) (2.32.5)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from transformers) (0.22.2)
Requirement already satisfied: safetensors>=0.4.3 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from transformers) (0.7.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from huggingface-hub>=0.20.0->sentence_transformers) (1.2.0)
Requirement already satisfied: python-terrier>=0.13.0 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from pyterrier-alpha) (1.0)
Requirement already satisfied: anyio<5,>=3.5.0 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from openai) (4.12.0)
Requirement already satisfied: distro<2,>=1.7.0 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from openai) (1.9.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from openai) (0.27.2)
Requirement already satisfied: jiter<1,>=0.10.0 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from openai) (0.12.0)
Requirement already satisfied: pydantic<3,>=1.9.0 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from openai) (2.12.5)
Requirement already satisfied: sniffio in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from openai) (1.3.1)
Requirement already satisfied: idna>=2.8 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from anyio<5,>=3.5.0->openai) (3.11)
Requirement already satisfied: certifi in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from httpx<1,>=0.23.0->openai) (2026.1.4)
Requirement already satisfied: httpcore==1.* in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from httpx<1,>=0.23.0->openai) (1.0.9)
Requirement already satisfied: h11>=0.16 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from httpcore==1.*->httpx<1,>=0.23.0->openai) (0.16.0)
Requirement already satisfied: annotated-types>=0.6.0 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from pydantic<3,>=1.9.0->openai) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.5 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from pydantic<3,>=1.9.0->openai) (2.41.5)
Requirement already satisfied: typing-inspection>=0.4.2 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from pydantic<3,>=1.9.0->openai) (0.4.2)
Requirement already satisfied: arrow in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from codecarbon) (1.4.0)
Requirement already satisfied: click in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from codecarbon) (8.3.1)
Requirement already satisfied: fief-client[cli] in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from codecarbon) (0.20.0)
Requirement already satisfied: prometheus_client in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from codecarbon) (0.23.1)
Requirement already satisfied: psutil>=6.0.0 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from codecarbon) (7.2.1)
Requirement already satisfied: py-cpuinfo in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from codecarbon) (9.0.0)
Requirement already satisfied: nvidia-ml-py in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from codecarbon) (13.590.44)
Requirement already satisfied: rapidfuzz in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from codecarbon) (3.14.3)
Requirement already satisfied: questionary in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from codecarbon) (2.1.1)
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Requirement already satisfied: typer in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from codecarbon) (0.21.1)
Requirement already satisfied: pyarrow>=21.0.0 in /home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages (from datasets>=3.0.0->lightning-ir) (22.0.0)
```

```
Requirement already satisfied: dill<0.4.1,>=0.3.0 in /home/kpch/miniconda3/envs/py312/lib
/python3.12/site-packages (from datasets>=3.0.0->lightning-ir) (0.4.0)
Requirement already satisfied: xxhash in /home/kpch/miniconda3/envs/py312/lib/python3.12/
site-packages (from datasets>=3.0.0->lightning-ir) (3.6.0)
Requirement already satisfied: multiprocess<0.70.19 in /home/kpch/miniconda3/envs/py312/l
ib/python3.12/site-packages (from datasets>=3.0.0->lightning-ir) (0.70.18)
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in /home/kpch/miniconda3/envs/p
y312/lib/python3.12/site-packages (from fsspec[http]<=2025.10.0,>=2023.1.0->datasets>=3.0
.0->lightning-ir) (3.13.3)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /home/kpch/miniconda3/envs/py31
2/lib/python3.12/site-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2025.10.0,
>=2023.1.0->datasets>=3.0.0->lightning-ir) (2.6.1)
Requirement already satisfied: aiosignal>=1.4.0 in /home/kpch/miniconda3/envs/py312/lib/p
ython3.12/site-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2025.10.0,>=2023.
1.0->datasets>=3.0.0->lightning-ir) (1.4.0)
Requirement already satisfied: attrs>=17.3.0 in /home/kpch/miniconda3/envs/py312/lib/pyth
on3.12/site-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2025.10.0,>=2023.1.0
->datasets>=3.0.0->lightning-ir) (25.4.0)
Requirement already satisfied: frozenlist>=1.1.1 in /home/kpch/miniconda3/envs/py312/lib/
python3.12/site-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2025.10.0,>=2023
.1.0->datasets>=3.0.0->lightning-ir) (1.8.0)
Requirement already satisfied: multidict<7.0,>=4.5 in /home/kpch/miniconda3/envs/py312/li
b/python3.12/site-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2025.10.0,>=20
23.1.0->datasets>=3.0.0->lightning-ir) (6.7.0)
Requirement already satisfied: propcache>=0.2.0 in /home/kpch/miniconda3/envs/py312/lib/p
ython3.12/site-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2025.10.0,>=2023.
1.0->datasets>=3.0.0->lightning-ir) (0.4.1)
Requirement already satisfied: yarl<2.0,>=1.17.0 in /home/kpch/miniconda3/envs/py312/lib/
python3.12/site-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2025.10.0,>=2023
.1.0->datasets>=3.0.0->lightning-ir) (1.22.0)
Requirement already satisfied: docstring-parser>=0.17 in /home/kpch/miniconda3/envs/py312
/lib/python3.12/site-packages (from jsonargparse[signatures]>=4.27.7->lightning-ir) (0.17
.0)
Requirement already satisfied: typeshed-client>=2.8.2 in /home/kpch/miniconda3/envs/py312
/lib/python3.12/site-packages (from jsonargparse[signatures]>=4.27.7->lightning-ir) (2.8.
2)
Requirement already satisfied: pyterrier>=1.0 in /home/kpch/miniconda3/envs/py312/lib/pyt
hon3.12/site-packages (from pyterrier[all]>=1.0->python-terrier>=0.13.0->pyterrier-alpha)
(1.0.1)
Requirement already satisfied: more_itertools in /home/kpch/miniconda3/envs/py312/lib/pyt
hon3.12/site-packages (from pyterrier>=1.0->pyterrier[all]>=1.0->python-terrier>=0.13.0->
pyterrier-alpha) (10.8.0)
Requirement already satisfied: deprecated in /home/kpch/miniconda3/envs/py312/lib/python3
.12/site-packages (from pyterrier>=1.0->pyterrier[all]>=1.0->python-terrier>=0.13.0->pyte
rrier-alpha) (1.3.1)
Requirement already satisfied: pytrec_eval_terrier>=0.5.3 in /home/kpch/miniconda3/envs/p
y312/lib/python3.12/site-packages (from pyterrier>=1.0->pyterrier[all]>=1.0->python-terri
er>=0.13.0->pyterrier-alpha) (0.5.10)
Requirement already satisfied: statsmodels in /home/kpch/miniconda3/envs/py312/lib/python
3.12/site-packages (from pyterrier>=1.0->pyterrier[all]>=1.0->python-terrier>=0.13.0->pyt
errier-alpha) (0.14.6)
Requirement already satisfied: lz4 in /home/kpch/miniconda3/envs/py312/lib/python3.12/sit
e-packages (from pyterrier>=1.0->pyterrier[all]>=1.0->python-terrier>=0.13.0->pyterrier-a
lpha) (4.4.5)
Requirement already satisfied: beautifulsoup4 in /home/kpch/miniconda3/envs/py312/lib/pyt
hon3.12/site-packages (from pyterrier>=1.0->pyterrier[all]>=1.0->python-terrier>=0.13.0->
pyterrier-alpha) (4.14.3)
Requirement already satisfied: inscriptis>=2.2.0 in /home/kpch/miniconda3/envs/py312/lib/
python3.12/site-packages (from ir_datasets->lightning-ir) (2.7.0)
Requirement already satisfied: lxml>=4.5.2 in /home/kpch/miniconda3/envs/py312/lib/python
3.12/site-packages (from ir_datasets->lightning-ir) (5.4.0)
Requirement already satisfied: trec-car-tools>=2.5.4 in /home/kpch/miniconda3/envs/py312/
lib/python3.12/site-packages (from ir_datasets->lightning-ir) (2.6)
Requirement already satisfied: warc3-wet>=0.2.3 in /home/kpch/miniconda3/envs/py312/lib/p
ython3.12/site-packages (from ir_datasets->lightning-ir) (0.2.5)
Requirement already satisfied: warc3-wet-clueweb09>=0.2.5 in /home/kpch/miniconda3/envs/p
y312/lib/python3.12/site-packages (from ir_datasets->lightning-ir) (0.2.5)
Requirement already satisfied: zlib-state>=0.1.3 in /home/kpch/miniconda3/envs/py312/lib/
python3.12/site-packages (from ir_datasets->lightning-ir) (0.1.10)
Requirement already satisfied: ijson>=3.1.3 in /home/kpch/miniconda3/envs/py312/lib/pytho
n3.12/site-packages (from ir_datasets->lightning-ir) (3.4.0.post0)
Requirement already satisfied: unlzw3>=0.2.1 in /home/kpch/miniconda3/envs/py312/lib/pyth
```

```
on3.12/site-packages (from ir_datasets->lightning-ir) (0.2.3)
Requirement already satisfied: soupsieve>=1.6.1 in /home/kpch/miniconda3/envs/py312/lib/p
ython3.12/site-packages (from beautifulsoup4->pyterrier>=1.0->pyterrier[all]>=1.0->python
-terrier>=0.13.0->pyterrier-alpha) (2.8.1)
Requirement already satisfied: charset_normalizer<4,>=2 in /home/kpch/miniconda3/envs/py3
12/lib/python3.12/site-packages (from requests->transformers) (3.4.4)
Requirement already satisfied: urllib3<3,>=1.21.1 in /home/kpch/miniconda3/envs/py312/lib
/python3.12/site-packages (from requests->transformers) (2.6.2)
Requirement already satisfied: pyjnius>=1.4.2 in /home/kpch/miniconda3/envs/py312/lib/pyt
hon3.12/site-packages (from pyterrier[all]>=1.0->python-terrier>=0.13.0->pyterrier-alpha)
(1.7.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /home/kpch/miniconda3/envs/py312/lib
/python3.12/site-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: accelerate>=0.26.0 in /home/kpch/miniconda3/envs/py312/lib
/python3.12/site-packages (from transformers[torch]>=4.50.0->lightning-ir) (1.12.0)
Requirement already satisfied: cbor>=1.0.0 in /home/kpch/miniconda3/envs/py312/lib/python
3.12/site-packages (from trec-car-tools>=2.5.4->ir_datasets->lightning-ir) (1.0.0)
Requirement already satisfied: importlib_resources>=1.4.0 in /home/kpch/miniconda3/envs/p
y312/lib/python3.12/site-packages (from typeshed-client>=2.8.2->jsonargparse[signatures]>
=4.27.7->lightning-ir) (6.5.2)
Requirement already satisfied: python-dateutil>=2.7.0 in /home/kpch/miniconda3/envs/py312
/lib/python3.12/site-packages (from arrow->codecarbon) (2.9.0.post0)
Requirement already satisfied: tzdata in /home/kpch/miniconda3/envs/py312/lib/python3.12/
site-packages (from arrow->codecarbon) (2025.3)
Requirement already satisfied: six>=1.5 in /home/kpch/miniconda3/envs/py312/lib/python3.1
2/site-packages (from python-dateutil>=2.7.0->arrow->codecarbon) (1.17.0)
Requirement already satisfied: wrapt<3,>=1.10 in /home/kpch/miniconda3/envs/py312/lib/pyt
hon3.12/site-packages (from deprecated->pyterrier>=1.0->pyterrier[all]>=1.0->python-terri
er>=0.13.0->pyterrier-alpha) (2.0.1)
Requirement already satisfied: jwcrypto<2.0.0,>=1.4 in /home/kpch/miniconda3/envs/py312/l
ib/python3.12/site-packages (from fief-client[cli]->codecarbon) (1.5.6)
Requirement already satisfied: yaspin in /home/kpch/miniconda3/envs/py312/lib/python3.12/
site-packages (from fief-client[cli]->codecarbon) (3.4.0)
Requirement already satisfied: cryptography>=3.4 in /home/kpch/miniconda3/envs/py312/lib/
python3.12/site-packages (from jwcrypto<2.0.0,>=1.4->fief-client[cli]->codecarbon) (46.0.
3)
Requirement already satisfied: cffi>=2.0.0 in /home/kpch/miniconda3/envs/py312/lib/python
3.12/site-packages (from cryptography>=3.4->jwcrypto<2.0.0,>=1.4->fief-client[cli]->codec
arbon) (2.0.0)
Requirement already satisfied: pycparser in /home/kpch/miniconda3/envs/py312/lib/python3.
12/site-packages (from cffi>=2.0.0->cryptography>=3.4->jwcrypto<2.0.0,>=1.4->fief-client[
cli]->codecarbon) (2.23)
Requirement already satisfied: MarkupSafe>=2.0 in /home/kpch/miniconda3/envs/py312/lib/py
thon3.12/site-packages (from jinja2->torch) (3.0.3)
Requirement already satisfied: lightning-utilities<2.0,>=0.10.0 in /home/kpch/miniconda3/
envs/py312/lib/python3.12/site-packages (from lightning->lightning-ir) (0.15.2)
Requirement already satisfied: torchmetrics<3.0,>0.7.0 in /home/kpch/miniconda3/envs/py31
2/lib/python3.12/site-packages (from lightning->lightning-ir) (1.8.2)
Requirement already satisfied: pytorch-lightning in /home/kpch/miniconda3/envs/py312/lib/
python3.12/site-packages (from lightning->lightning-ir) (2.6.0)
Requirement already satisfied: pytz>=2020.1 in /home/kpch/miniconda3/envs/py312/lib/pytho
n3.12/site-packages (from pandas->lightning-ir) (2025.2)
Requirement already satisfied: prompt_toolkit<4.0,>=2.0 in /home/kpch/miniconda3/envs/py3
12/lib/python3.12/site-packages (from questionary->codecarbon) (3.0.52)
Requirement already satisfied: wcwidth in /home/kpch/miniconda3/envs/py312/lib/python3.12
/site-packages (from prompt_toolkit<4.0,>=2.0->questionary->codecarbon) (0.2.14)
Requirement already satisfied: markdown-it-py>=2.2.0 in /home/kpch/miniconda3/envs/py312/
lib/python3.12/site-packages (from rich->codecarbon) (4.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /home/kpch/miniconda3/envs/py31
2/lib/python3.12/site-packages (from rich->codecarbon) (2.19.2)
Requirement already satisfied: mdurl~=0.1 in /home/kpch/miniconda3/envs/py312/lib/python3
.12/site-packages (from markdown-it-py>=2.2.0->rich->codecarbon) (0.1.2)
Requirement already satisfied: joblib>=1.3.0 in /home/kpch/miniconda3/envs/py312/lib/pyth
on3.12/site-packages (from scikit-learn->sentence_transformers) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.2.0 in /home/kpch/miniconda3/envs/py312/l
ib/python3.12/site-packages (from scikit-learn->sentence_transformers) (3.6.0)
Requirement already satisfied: patsy>=0.5.6 in /home/kpch/miniconda3/envs/py312/lib/pytho
n3.12/site-packages (from statsmodels->pyterrier>=1.0->pyterrier[all]>=1.0->python-terrie
r>=0.13.0->pyterrier-alpha) (1.0.2)
Requirement already satisfied: shellingham>=1.3.0 in /home/kpch/miniconda3/envs/py312/lib
/python3.12/site-packages (from typer->codecarbon) (1.5.4)
Requirement already satisfied: termcolor<4.0,>=3.2 in /home/kpch/miniconda3/envs/py312/li
```

```
b/python3.12/site-packages (from yaspin->fief-client[cli]->codecarbon) (3.3.0)
```

In [26]:

```
# # Install OpenJDK from conda-forge
# !conda install -c conda-forge openjdk -y
```

In [2]:

```
from tira.ir_datasets_util import register_dataset
from pyterrier.datasets import get_dataset

# On the bwJupyter cluster, this file exists in this shared path.
# Please, do not publish this data openly if you must download it.
register_dataset("__shared/clueweb12.zip", "clueweb12-subsample")
dataset = get_dataset("irds:clueweb12-subsample")
```

Implementation

Use as many cells as you like below to implement your model on the dataset provided above. Once you've implemented your model compare it to BM25 and the pipeline you created in project Part 1. You should use the evaluation measures listed on the leaderboard system to evaluate your systems.

In [3]:

```
# Your solution here
```

In [4]:

```
import os
import datetime
import torch
import pandas as pd
import pyterrier as pt
from pathlib import Path
from pyterrier.measures import nDCG, P, RR
from ir_measures import nDCG, RR, P

from lightning_ir_pyterrier import BiEncoder, CrossEncoder
from lightning_ir import TorchDenseIndexConfig, TorchDenseSearchConfig
from pyterrier.text import get_text
import pyterrier_alpha as pta


device = "cpu"
if torch.cuda.is_available():
    device = "cuda"
if torch.backends.mps.is_available():
    device = "mps"
```

In [4]:

```
for doc in dataset.get_corpus_iter():
    # print("Raw Object:", doc)
    # Get just the field names
    if hasattr(doc, '_fields'):
        print("Fields available:", doc._fields)
    else:
        # Fallback if it's a standard dictionary
        print("Fields available:", doc.keys())
    break
```

```
Fields available: dict_keys(['text', 'docno'])
```

Comparison to a BM25 baseline and your own model from Part 1

In [5]:

```
if not pt.java.started():
    pt.init()

# Updated Cascade Model from Part 1:
K_CANDIDATES = 1000       # BM25 retrieves 1000

# Indexer
pt_index_path = "./index_clueweb12-subsample"
if not os.path.exists(pt_index_path + "/data.properties"):
  indexer = pt.index.IterDictIndexer(
      pt_index_path,
      meta_reverse=[],
      fields=["text"],
      meta={'docno': 25}
  )
  index_ref = indexer.index(dataset.get_corpus_iter())
else:
  index_ref = pt_index_path
index = pt.IndexFactory.of(index_ref)

# Baseline Model: truncated to the same cut-off of K_CANDIDATES for better evaluation ana
lysis
bm25_baseline = (pt.terrier.Retriever(index, wmodel="BM25") % K_CANDIDATES)

# Simple Cross Encoder
cross_encoder = CrossEncoder(
    model_name_or_path="webis/monoelectra-base",
    batch_size=1,
    verbose=True,
    use_gpu=torch.cuda.is_available()
)
cross_encoder._module.to(device)
cross_encoder._module.device

# Cascade Pipeline (BM25 >> cross_encoder)
# Stage 1: BM25 retrieves K_CANDIDATES candidates
bm25_stage1 = bm25_baseline

# Stage 2: Cross-Encoder re-ranks the K_CANDIDATES
cascade_model = (
    bm25_stage1
    >> get_text(dataset)
    >> cross_encoder
)
```

```
Java started and loaded: pyterrier.java.colab, pyterrier.java, pyterrier.java.24, pyterri
er.terrier.java [version=5.11 (build: craig.macdonald 2025-01-13 21:29), helper_version=0
.0.8]
/tmp/ipykernel_3435548/3334950383.py:2: DeprecationWarning: Call to deprecated method pt.
init(). Deprecated since version 0.11.0.
java is now started automatically with default settings. To force initialisation early, r
un:
pt.java.init() # optional, forces java initialisation
  pt.init()
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
The tokenizer class you load from this checkpoint is not the same type as the class this
function is called from. It may result in unexpected tokenization.
The tokenizer class you load from this checkpoint is 'ElectraTokenizer'.
The class this function is called from is 'CrossEncoderElectraTokenizerFast'.
```

In [12]:

```
# nDCG@k: normalised discounted cumulative gain
# P@k: precision at depth k, fraction of top-k are relevant
```

```
# RR: Rank of the first relevant item
topics = dataset.get_topics(variant='text')
print(f"Running experiment on {len(topics)} topics, evaluating at k={K_CANDIDATES}...")
experiment_results = pt.Experiment(
    [
        bm25_baseline,
        cascade_model
    ],
    dataset.get_topics(variant='text'),
    dataset.get_qrels(),
    eval_metrics=[nDCG@10, nDCG@100, RR, P@10],
    names=["BM25 Baseline", "Cascade Model from Part 1 (BM25 >> Cross-Encoder)"],
    verbose=True,
)
print(experiment_results)
```

```
Running experiment on 50 topics, evaluating at k=1000...
```

```
/home/kpch/miniconda3/envs/py312/lib/python3.12/site-packages/pyterrier/_evaluation/_validation.py:43: UserWarning: Transformer Cascade Model from Part 1 (BM25 >> Cross-Encoder) ((TerrierRetr(BM25) >> RankCutoff(1000) >> <pyterrier.datasets._irds.IRDSTextLoader object at 0x7e0a695caed0> >> <lightning_ir_pyterrier.CrossEncoder object at 0x7e0a695cb800>)) at position 1 failed to validate: Cannot determine outputs for (TerrierRetr(BM25) >> RankCutoff(1000) >> <pyterrier.datasets._irds.IRDSTextLoader object at 0x7e0a695caed0> >> <lightning_ir_pyterrier.CrossEncoder object at 0x7e0a695cb800>) with inputs: ['qid', 'query'] - if your pipeline works, set validate='ignore' to remove this warning, or add transform_output method to the transformers in this pipeline to clarify how it works. See https://pyterrier.readthedocs.io/en/latest/troubleshooting/inspection.html for more information.
  warn(
```

```
                                                name        RR   P@10  \
0                                   BM25 Baseline  0.550403  0.504
1  Cascade Model from Part 1 (BM25 >> Cross-Encoder)  0.587185  0.350

    nDCG@10  nDCG@100
0  0.306496  0.362850
1  0.254421  0.294331
```

Retrieval pipeline for web documents

In [11]:

```
# Calculate lengths for all documents in the corpus
# We split by whitespace to count words
doc_lengths = [len(doc['text'].split()) for doc in dataset.get_corpus_iter()]

if doc_lengths:
    avg_len = sum(doc_lengths) / len(doc_lengths)
    max_len = max(doc_lengths)
    min_len = min(doc_lengths)
else:
    avg_len = 0
    max_len = 0
    min_len = 0

print(f"Average Document Length: {avg_len:.2f} words")
print(f"Maximum Document Length: {max_len} words")
print(f"Minimum Document Length: {min_len} words")
```

```
Average Document Length: 2110.74 words
Maximum Document Length: 335925 words
Minimum Document Length: 0 words
```

In [6]:

```
# Initialize PyTerrier (Java Check)
if not pt.java.started():
    pt.init()
```

In [26]:

```
# Check if we have raw HTML tags
for doc in dataset.get_corpus_iter():
    print(doc["text"][:500])
    break
```

```
Louis Vuitton Damier Ebene Canvas Belts Stylish-0004,Cheap Louis Vuitton Belts,wholesale
Louis Vuitton Leather Belts Home Log In
Currencies:
Languages: English
Shopping Cart: now in your cart 0 items

Home
  • Return & Exchange
  • Shipping
  • Payment
  • About wholesale
  • Contact Us
  • Live chat
    • Louis Vuitton Bags
      Louis Vuitton 2010 New Style Louis Vuitton Damier Handbags Louis Vuitton Epi Leathe
r Louis Vuitton Leather Bags Louis Vuitton Men Handbags Louis Vuitton Monogram Hand
```

In [7]:

```
# BUILD INDEXES
def build_indexes(dataset, sparse_path, dense_path, bi_model_id, batch_size=32):
    # A. Sparse
    if not os.path.exists(sparse_path + "/data.properties"):
        print(f"--- Building Sparse Index at {sparse_path} ---")
        indexer = pt.IterDictIndexer(
            sparse_path,
            meta_reverse=['docno'],                    # Allows looking up docid intege
rs from string docnos
            meta={'docno': 25, 'text': 32000},         # Store up to ~32k characters so
the footer is actually saved
            blocks=True,                               # Enables phrase search ("query
terms")
            fields=['text']
        )
        indexer.index(dataset.get_corpus_iter())

    # B. Dense (Bi-Encoder)
    safe_model_name = bi_model_id.replace("/", "_")
    specific_dense_path = os.path.join(dense_path, safe_model_name)

    if not os.path.exists(specific_dense_path):
        print(f"--- Building Dense Index at {specific_dense_path} ---")
        bi_indexer = BiEncoder(
            model_name_or_path=bi_model_id,
            index_dir=specific_dense_path,
            index_config=TorchDenseIndexConfig(),
            text_field="text",
            batch_size=batch_size,
            use_gpu=True,
            verbose=True
        )
        if hasattr(bi_indexer, "_module"):
             bi_indexer._module.to("cuda" if torch.cuda.is_available() else "cpu")
        bi_indexer.index(dataset.get_corpus_iter())

    return specific_dense_path
```

In [8]:

```
def run_toggled_experiment(
    dataset,
    sparse_path,
    dense_index_path,
```

```
    topics_variant='text',

    # --- TOGGLES ---
    use_bm25f=False,         # Use Title weights?
    use_rm3=False,           # Use Query Expansion?
    use_bi_encoder=False,    # Use Dense Retrieval?
    use_rrf=False,           # Use Rank Fusion?
    use_cross_encoder=True, # Use Final Reranker?

    # --- PARAMETERS ---
    bi_model_id="webis/bert-bi-encoder",
    ce_model_id="webis/monoelectra-base",

    # if using bm25f, but out docs don't have a predefined 'title' and extracting doesn't
seem worth the effort
    bm25f_params={"w.title": 3.0, "w.text": 1.0},

    rm3_terms=10,
    rm3_docs=10,
    window_len=300,         # 300 tokens is robust for web docs
    window_stride=150,      # 50% overlap
    bi_top_k=1000,          # Rerank depth for Bi-Encoder
    ce_top_k=100,           # Rerank depth for Cross-Encoder

    batch_size=32
):

    index = pt.IndexFactory.of(sparse_path)
    device = "cuda" if torch.cuda.is_available() else "cpu"

    # --- STAGE 1: Sparse Retrieval ---
    if use_bm25f:
        print(" [1] Using BM25F (Fielded)")
        controls = {"wmodel": "BM25F"}
        controls.update(bm25f_params)
        pipeline = pt.terrier.Retriever(index, controls=controls)
        name = "BM25F"
    else:
        print(" [1] Using Standard BM25")
        pipeline = pt.terrier.Retriever(index, wmodel="BM25")
        name = "BM25"

    # --- STAGE 2: Query Expansion ---
    if use_rm3:
        print(f" [2] Adding RM3 ({rm3_terms} terms)")
        rm3 = pt.rewrite.RM3(index, fb_terms=rm3_terms, fb_docs=rm3_docs)
        pipeline = pipeline >> rm3 >> pipeline
        name += "+RM3"

    # --- STAGE 3: Dense Retrieval (Bi-Encoder) ---
    if use_bi_encoder:
        print(f" [3] Adding Bi-Encoder ({bi_model_id})")

        # 1. Define the Bi-Encoder
        bi_enc = BiEncoder(
            model_name_or_path=bi_model_id,
            index_dir=Path(dense_index_path),  # It needs the index to search!
            index_config=TorchDenseIndexConfig(),
            search_config=TorchDenseSearchConfig(k=bi_top_k),
            text_field="text",
            batch_size=batch_size,
            use_gpu=True
        )
        if hasattr(bi_enc, "_module"): bi_enc._module.to(device)

        # 2. Define the Branch (PARALLEL)
        # We do NOT put 'pipeline >>' in front.
        # We treat 'bi_enc' as a standalone retriever that starts from the user query.
        dense_pipe = bi_enc

        # 3. Apply Fusion
        if use_rrf:
```

```
            print("     -> Applying Fusion (RRF)")
            # pipeline = BM25 results
            # dense_pipe = Bi-Encoder results (New docs!)
            # RRFusion merges them together
            pipeline = pta.fusion.RRFusion(pipeline, dense_pipe, k=60)
            name += "+BiEnc+RRF"
        else:
            # If RRF is off, we strictly replace BM25 with Bi-Encoder
            pipeline = dense_pipe
            name = "BiEnc"

    # --- STAGE 4: Cross-Encoder ---
    if use_cross_encoder:
        print(f" [4] Adding Cross-Encoder ({ce_model_id})")
        ce = CrossEncoder(
            model_name_or_path=ce_model_id,
            text_field="text",
            batch_size=batch_size,
            use_gpu=True
        )
        if hasattr(ce, "_module"): ce._module.to(device)

        pipeline = (
            (pipeline % ce_top_k)
            >> pt.text.get_text(dataset, "text")
            >> pt.text.sliding(length=window_len, stride=window_stride, text_attr='text'
, prepend_attr=None) # otherwise looks for "body"
            >> ce
            >> pt.text.max_passage()
        )
        name += "+CE"

    # --- Execution ---
    print(f"\nRunning Pipeline: {name}")

    # We run .transform() first. This does the retrieval + cross-encoding
    start_time = datetime.datetime.now()
    run_out = pipeline.transform(dataset.get_topics(variant=topics_variant))
    print(f"Pipeline finished in {datetime.datetime.now() - start_time}")

    # Save results
    timestamp = datetime.datetime.now().strftime("%Y%m%d_%H%M")
    safe_name = name.replace("+", "_").replace("(", "").replace(")", "")
    pt.io.write_results(run_out, f"run_{timestamp}_{safe_name}.res")

    # Calculate Metrices
    metrics = [nDCG@10, nDCG@100, P@10, RR(rel=1)]
    exp = pt.Experiment(
        [run_out],  # <--- Pass the result dataframe, not the pipeline: PyTerrier sees i
t's already data and just computes the math
        dataset.get_topics(variant=topics_variant),
        dataset.get_qrels(),
        eval_metrics=metrics,
        names=[name],
        verbose=True,         # shows progress
        validate=False        # surpress warnings
    )
    exp.to_csv(f"results_{timestamp}_{safe_name}.csv")

    print(exp)
    return exp
```

In [9]:

```
# MAIN
SPARSE_PATH = "./indices/sparse_clueweb"
DENSE_ROOT  = "./indices/dense_clueweb"
BI_MODEL = "webis/bert-bi-encoder"
CE_MODEL = "webis/monoelectra-large"

# Step 1. Build Indexes
```

```
dense_path = build_indexes(dataset, SPARSE_PATH, DENSE_ROOT, BI_MODEL)
```

In [30]:

```
# Step 2. Run Experiment
run_toggled_experiment(
    dataset,
    SPARSE_PATH,
    dense_path,
    topics_variant='text',

    # --- TOGGLES ---
    use_bm25f=False,        # Strong title baseline
    use_rm3=False,          # Essential for short queries
    use_bi_encoder=True,    # Optional: Turn on for deep semantic search
    use_rrf=True,           # Optional: Only works if bi_encoder is True
    use_cross_encoder=True,# The precision booster

    # --- PARAMETERS ---
    bi_model_id=BI_MODEL,
    ce_model_id=CE_MODEL,
    # bm25f_params={"w.title": 5.0, "w.text": 1.0},
    # rm3_terms=10,
    # rm3_docs=10,
    window_len=300,         # 300 tokens is robust for web docs
    window_stride=150,      # 50% overlap
    bi_top_k=1000,          # Rerank depth for Bi-Encoder
    ce_top_k=100,           # Rerank depth for Cross-Encoder
)
```

```
 [1] Using Standard BM25
 [3] Adding Bi-Encoder (webis/bert-bi-encoder)
```

```
You are using a model of type lir-dpr to instantiate a model of type bert. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type lir-dpr to instantiate a model of type bert. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type lir-dpr to instantiate a model of type bert. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type lir-dpr to instantiate a model of type bert. This is not su
pported for all configurations of models and can yield errors.
The tokenizer class you load from this checkpoint is not the same type as the class this
function is called from. It may result in unexpected tokenization.
The tokenizer class you load from this checkpoint is 'BertTokenizer'.
The class this function is called from is 'BiEncoderBertTokenizerFast'.
```

```
     -> Applying Fusion (RRF)
 [4] Adding Cross-Encoder (webis/monoelectra-large)
```

```
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
The tokenizer class you load from this checkpoint is not the same type as the class this
function is called from. It may result in unexpected tokenization.
The tokenizer class you load from this checkpoint is 'ElectraTokenizer'.
The class this function is called from is 'CrossEncoderElectraTokenizerFast'.
```

```
Running Pipeline: BM25+BiEnc+RRF+CE
calling sliding on df of 5000 rows


Pipeline finished in 0:03:39.212369


                name        RR   P@10   nDCG@10  nDCG@100
0  BM25+BiEnc+RRF+CE  0.638627  0.428  0.315834  0.362112
```

Out[30]:

	name	RR	P@10	nDCG@10	nDCG@100
0	BM25+BiEnc+RRF+CE	0.638627	0.428	0.315834	0.362112

In [111]:

```
run_toggled_experiment(
    dataset,
    SPARSE_PATH,
    dense_path,
    topics_variant='text',

    # --- TOGGLES  ---
    use_bm25f=False,        # Strong title baseline
    use_rm3=False,          # Essential for short queries
    use_bi_encoder=True,    # Optional: Turn on for deep semantic search
    use_rrf=True,           # Optional: Only works if bi_encoder is True
    use_cross_encoder=True,# The precision booster

    # --- PARAMETERS ---
    bi_model_id=BI_MODEL,
    ce_model_id="webis/monoelectra-base",
    # bm25f_params={"w.title": 5.0, "w.text": 1.0},
    # rm3_terms=10,
    # rm3_docs=10,
    window_len=300,         # 300 tokens is robust for web docs
    window_stride=150,      # 50% overlap
    bi_top_k=1000,          # Rerank depth for Bi-Encoder
    ce_top_k=100,           # Rerank depth for Cross-Encoder
)
```

```
 [1] Using Standard BM25
 [3] Adding Bi-Encoder (webis/bert-bi-encoder)
```

```
You are using a model of type lir-dpr to instantiate a model of type bert. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type lir-dpr to instantiate a model of type bert. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type lir-dpr to instantiate a model of type bert. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type lir-dpr to instantiate a model of type bert. This is not su
pported for all configurations of models and can yield errors.
The tokenizer class you load from this checkpoint is not the same type as the class this
function is called from. It may result in unexpected tokenization.
The tokenizer class you load from this checkpoint is 'BertTokenizer'.
The class this function is called from is 'BiEncoderBertTokenizerFast'.
```

```
     -> Applying Fusion (RRF)
 [4] Adding Cross-Encoder (webis/monoelectra-base)
```

```
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
The tokenizer class you load from this checkpoint is not the same type as the class this
function is called from. It may result in unexpected tokenization.
The tokenizer class you load from this checkpoint is 'ElectraTokenizer'.
The class this function is called from is 'CrossEncoderElectraTokenizerFast'.
```

```
Running Pipeline: BM25+BiEnc+RRF+CE
calling sliding on df of 5000 rows


Pipeline finished in 0:01:32.867664


                name        RR   P@10   nDCG@10  nDCG@100
0  BM25+BiEnc+RRF+CE  0.624785  0.424  0.307404  0.359267
```

Out[111]:

	name	RR	P@10	nDCG@10	nDCG@100
0	BM25+BiEnc+RRF+CE	0.624785	0.424	0.307404	0.359267

In [112]:

```
run_toggled_experiment(
    dataset,
    SPARSE_PATH,
    dense_path,
    topics_variant='text',

    # --- TOGGLES ---
    use_bm25f=False,        # Strong title baseline
    use_rm3=False,          # Essential for short queries
    use_bi_encoder=False,   # Optional: Turn on for deep semantic search
    use_rrf=False,          # Optional: Only works if bi_encoder is True
    use_cross_encoder=True,# The precision booster

    # --- PARAMETERS ---
    bi_model_id=BI_MODEL,
    ce_model_id=CE_MODEL,
    # bm25f_params={"w.title": 5.0, "w.text": 1.0},
    # rm3_terms=10,
    # rm3_docs=10,
    window_len=300,         # 300 tokens is robust for web docs
    window_stride=150,      # 50% overlap
    bi_top_k=1000,          # Rerank depth for Bi-Encoder
    ce_top_k=100,           # Rerank depth for Cross-Encoder
)
```

```
 [1] Using Standard BM25
 [4] Adding Cross-Encoder (webis/monoelectra-large)
```

```
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
The tokenizer class you load from this checkpoint is not the same type as the class this
function is called from. It may result in unexpected tokenization.
The tokenizer class you load from this checkpoint is 'ElectraTokenizer'.
The class this function is called from is 'CrossEncoderElectraTokenizerFast'.
```

```
Running Pipeline: BM25+CE
calling sliding on df of 5000 rows


Pipeline finished in 0:03:56.058426


      name         RR    P@10   nDCG@10  nDCG@100
0  BM25+CE  0.718799   0.526  0.374639  0.390035
```

Out[112]:

	name	RR	P@10	nDCG@10	nDCG@100
0	BM25+CE	0.718799	0.526	0.374639	0.390035

In [115]:

```
run_toggled_experiment(
    dataset,
    SPARSE_PATH,
    dense_path,
    topics_variant='text',
```

```
    # --- TOGGLES ---
    use_bm25f=False,        # Strong title baseline
    use_rm3=True,           # Essential for short queries
    use_bi_encoder=True,    # Optional: Turn on for deep semantic search
    use_rrf=True,           # Optional: Only works if bi_encoder is True
    use_cross_encoder=True,# The precision booster

    # --- PARAMETERS ---
    bi_model_id=BI_MODEL,
    ce_model_id=CE_MODEL,
    rm3_terms=12,
    rm3_docs=12,
    window_len=300,         # 300 tokens is robust for web docs
    window_stride=150,      # 50% overlap
    bi_top_k=1000,          # Rerank depth for Bi-Encoder
    ce_top_k=100,           # Rerank depth for Cross-Encoder
)
```

```
 [1] Using Standard BM25
 [2] Adding RM3 (12 terms)
 [3] Adding Bi-Encoder (webis/bert-bi-encoder)
```

```
You are using a model of type lir-dpr to instantiate a model of type bert. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type lir-dpr to instantiate a model of type bert. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type lir-dpr to instantiate a model of type bert. This is not su
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You are using a model of type lir-dpr to instantiate a model of type bert. This is not su
pported for all configurations of models and can yield errors.
The tokenizer class you load from this checkpoint is not the same type as the class this
function is called from. It may result in unexpected tokenization.
The tokenizer class you load from this checkpoint is 'BertTokenizer'.
The class this function is called from is 'BiEncoderBertTokenizerFast'.
```

```
     -> Applying Fusion (RRF)
 [4] Adding Cross-Encoder (webis/monoelectra-large)
```

```
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
The tokenizer class you load from this checkpoint is not the same type as the class this
function is called from. It may result in unexpected tokenization.
The tokenizer class you load from this checkpoint is 'ElectraTokenizer'.
The class this function is called from is 'CrossEncoderElectraTokenizerFast'.
```

```
Running Pipeline: BM25+RM3+BiEnc+RRF+CE
calling sliding on df of 5000 rows


Pipeline finished in 0:04:09.751001

                    name        RR  P@10   nDCG@10  nDCG@100
0  BM25+RM3+BiEnc+RRF+CE  0.591659  0.39  0.264471  0.302612
```

Out[115]:

	name	RR	P@10	nDCG@10	nDCG@100
0	**BM25+RM3+BiEnc+RRF+CE**	**0.591659**	**0.39**	**0.264471**	**0.302612**

In [33]:

```
run_toggled_experiment(
    dataset,
    SPARSE_PATH,
    dense_path,
```

```
    topics_variant='text',

    # --- TOGGLES ---
    use_bm25f=False,         # Strong title baseline
    use_rm3=False,           # Essential for short queries
    use_bi_encoder=True,     # Optional: Turn on for deep semantic search
    use_rrf=False,            # Optional: Only works if bi_encoder is True
    use_cross_encoder=False,# The precision booster

    # --- PARAMETERS ---
    bi_model_id=BI_MODEL,
    window_len=300,         # 300 tokens is robust for web docs
    window_stride=150,      # 50% overlap
    bi_top_k=1000,          # Rerank depth for Bi-Encoder
    ce_top_k=100,           # Rerank depth for Cross-Encoder
)
```

```
 [1] Using Standard BM25
 [3] Adding Bi-Encoder (webis/bert-bi-encoder)
```

```
You are using a model of type lir-dpr to instantiate a model of type bert. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type lir-dpr to instantiate a model of type bert. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type lir-dpr to instantiate a model of type bert. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type lir-dpr to instantiate a model of type bert. This is not su
pported for all configurations of models and can yield errors.
The tokenizer class you load from this checkpoint is not the same type as the class this
function is called from. It may result in unexpected tokenization.
The tokenizer class you load from this checkpoint is 'BertTokenizer'.
The class this function is called from is 'BiEncoderBertTokenizerFast'.
```

```
Running Pipeline: BiEnc
Pipeline finished in 0:00:00.135730
```

```
    name        RR    P@10   nDCG@10  nDCG@100
0  BiEnc  0.523022   0.314  0.229803  0.250048
```

Out[33]:

	name	RR	P@10	nDCG@10	nDCG@100
0	**BiEnc**	**0.523022**	**0.314**	**0.229803**	**0.250048**

From these experiments we can see that:

- **BM25+CE with parameters window=300, stride=150, ce_model="webis/monoelectra-large" performs the best for the given dataset**
- **Query Expansion via RM3 doesn't aid at all as it may be adding irrelevant terms**
- **Using Bi-enocder as a parallel retrieval and combining with BM25 also is not fruitful**

Let's trying fusing a model of BM25+CE and Bi_encoder

In [66]:

```
def rrf_fuse_files(file_path_1, file_path_2, k=60):
    print(f"Fusing:\n A: {file_path_1}\n B: {file_path_2}")

    res_1 = pt.io.read_results(file_path_1)
    res_2 = pt.io.read_results(file_path_2)

    # 2. Add RRF Scores
    # Score = 1 / (k + rank)
    res_1['rrf_score'] = 1 / (k + res_1['rank'])
    res_2['rrf_score'] = 1 / (k + res_2['rank'])

    # We merge the two dataframes
```

```
    combined = pd.concat([res_1[['qid', 'docno', 'rrf_score']],
                          res_2[['qid', 'docno', 'rrf_score']]])

    # Sum scores for the same (qid, docno) pair
    fused = combined.groupby(['qid', 'docno'])['rrf_score'].sum().reset_index()

    fused = fused.rename(columns={'rrf_score': 'score'})
    # Sort by score descending
    fused = fused.sort_values(['qid', 'score'], ascending=[True, False])
    # Recalculate rank
    fused['rank'] = fused.groupby('qid').cumcount()

    return fused

# Ensemble with BM25+CE and BI_ENC
file_bm25_ce = "run_20260109_0010_BM25_CE.res"
file_bi_enc  = "run_20260109_0311_BiEnc.res"
```

In [46]:

```
# Fuse them!
fused_run = rrf_fuse_files(file_bm25_ce, file_bi_enc, k=5)

# 4. Evaluate the Ensemble
metrics = [nDCG@10, nDCG@100, P@10, RR(rel=1)]
pt.Experiment(
    [fused_run],
    dataset.get_topics(variant='text'),
    dataset.get_qrels(),
    eval_metrics=metrics,
    names=["Ensemble"]
)
```

```
Fusing:
 A: run_20260109_0010_BM25_CE.res
 B: run_20260109_0311_BiEnc.res
```

Out[46]:

	name	RR	P@10	nDCG@10	nDCG@100
0	Ensemble	0.708953	0.454	0.341521	0.374339

Since our Bi_Encoder is bad than BM25+CE, it doesn't help in fusing them.

LLM Query Expansion

In [8]:

```
import pandas as pd
from openai import OpenAI
from tqdm import tqdm
from secret import GWDG_API_KEY
```

In [9]:

```
def generate_hyde_gwdg(
    dataset,
    api_key,
    variant='text',
    model_name="meta-llama-3.1-70b-instruct",
    system_prompt="You are a helpful assistant. Write a concise, factual passage that ans
wers the user's question directly.",
    prompt_template="Question: {query}\nPassage:",
    temperature=0.7,
    max_tokens=150
):
    print(f"--- Connecting to GWDG Chat AI ({model_name}) ---")
    client = OpenAI(
```

```
        base_url="https://chat-ai.academiccloud.de/v1",
        api_key=api_key
    )

    topics = dataset.get_topics(variant=variant).copy()
    new_queries = []

    print(f"--- Generating Hypothetical Documents (Temp={temperature}) ---")
    for q in tqdm(topics['query'], desc="Hallucinating"):
        try:
            # Format the user message dynamically
            user_content = prompt_template.format(query=q)

            response = client.chat.completions.create(
                model=model_name,
                messages=[
                    {"role": "system", "content": system_prompt},
                    {"role": "user", "content": user_content}
                ],
                temperature=temperature,
                max_tokens=max_tokens
            )

            fake_doc = response.choices[0].message.content.strip()
            fake_doc = fake_doc.replace("\n", " ") # Flatten for BM25

            # Combine: Query + Hallucination
            combined_query = f"{q} {fake_doc}"
            new_queries.append(combined_query)

        except Exception as e:
            print(f"Error on query '{q}': {e}")
            new_queries.append(q) # Fallback

    topics['query_original'] = topics['query']
    topics['query'] = new_queries
    print(topics['query_original'])
    print(topics['query'])

    return topics
```

In [10]:

```
def run_hyde_experiment(dataset, custom_topics, sparse_path, ce_model_id, batch_size=32)
:
    # 1. Setup BM25 (The Retriever)
    index = pt.IndexFactory.of(sparse_path)
    # We rely on BM25 to match the "fake answer" to "real documents"
    retriever = pt.terrier.Retriever(index, wmodel="BM25")

    # 2. Setup Cross-Encoder (The Filter)
    ce = CrossEncoder(
        model_name_or_path=ce_model_id,
        text_field="text",
        batch_size=max(1, batch_size // 2),
        use_gpu=True
    )
    if hasattr(ce, "_module"): ce._module.to("cuda")

    # 3. Build Pipeline
    # Note: We do NOT use RM3 here because the LLM already expanded the query
    pipeline = (
        (retriever % 100) # Retrieve top 100 using HyDE query
        >> pt.text.get_text(dataset, "text")
        >> pt.text.sliding(length=300, stride=150, text_attr='text', prepend_attr=None)
        >> ce
        >> pt.text.max_passage()
    )

    name = "BM25_HyDE_CE_Large"
    print(f"\nRunning Pipeline: {name}")
```

```
    # 4. EXECUTE with CUSTOM TOPICS
    start_time = datetime.datetime.now()
    run_out = pipeline.transform(custom_topics)
    print(f"Pipeline finished in {datetime.datetime.now() - start_time}")

    # 5. Evaluate
    metrics = [nDCG@10, nDCG@100, P@10, RR(rel=1)]
    exp = pt.Experiment(
        [run_out],
        custom_topics,        # Pass custom topics here too for ID matching
        dataset.get_qrels(),
        eval_metrics=metrics,
        names=[name],
        verbose=True,
        validate=False
    )

    timestamp = datetime.datetime.now().strftime("%Y%m%d_%H%M")
    pt.io.write_results(run_out, f"run_{timestamp}_{name}.res")
    print(exp)
    return exp
```

In [14]:

```
SPARSE_PATH = "./indices/sparse_clueweb"
DENSE_ROOT  = "./indices/dense_clueweb"
BI_MODEL = "webis/bert-bi-encoder"
CE_MODEL = "webis/monoelectra-large"

dense_path = build_indexes(dataset, SPARSE_PATH, DENSE_ROOT, BI_MODEL)

hyde_topics_keywords = generate_hyde_gwdg(
    dataset,
    api_key=GWDG_API_KEY,
    model_name="meta-llama-3.1-70b-instruct",
    system_prompt="You are a helpful assistant. Write a concise, factual passage that ans
wers the user's question directly.",
    prompt_template="Question: {query}\nPassage:",
    temperature=0.7,
    max_tokens=150
)

# Run the pipeline
run_hyde_experiment(
    dataset,
    hyde_topics_keywords,
    SPARSE_PATH,
    "webis/monoelectra-large"
)
```

```
--- Connecting to GWDG Chat AI (meta-llama-3.1-70b-instruct) ---
--- Generating Hypothetical Documents (Temp=0.7) ---
Hallucinating: 100%|████████████████████████████████████████| 50/50 [03:13<00:00,  3.87s/it
]
0                                identifying spider bites
1                                 history of orcas island
2                                           tooth abscess
3                                barrett's esophagus
4                                           teddy bears
5                          patron saint of mental illness
6                                   holes by louis sachar
7                                               hip roof
8                                           carpenter bee
9                              the american revolutionary
10                              folk remedies sore throat
11                                          balding cure
12                                evidence for evolution
13                       tribe formerly living in alabama
14                                          F5 tornado
15                                  symptoms of heart attack
```

```
15                                symptoms of heart attack
16                                    feliz navidad lyrics
17                                     benefits of running
18                                  marshall county schools
19                                                 sun tzu
20                halloween activities for middle school
21                                   dreams interpretation
22                                        wilson's disease
23                                        golf instruction
24                                                 uss cole
25     how has african american music influence history
26                                         bewitched cast
27                                          mister rogers
28                                            game theory
29                                 view my internet history
30                                         ketogenic diet
31                           nasa interplanetary missions
32                                         hayrides in pa
33                          where to find morel mushrooms
34                                   magnesium rich foods
35                              common schizophrenia drugs
36                  carotid cavernous fistula treatment
37                                          fidel castro
38                                        benefits of yoga
39                                           norway spruce
40                                sangre de cristo mountains
41              history of the electronic medical record
42    educational advantages of social networking sites
43                                        flowering plants
44                               how to tie a windsor knot
45                           recycling lead acid batteries
46                                        altitude sickness
47                  medical care and jehovah's witnesses
48                                pink slime in ground beef
49                                     how to find the mean
Name: query_original, dtype: object
0     identifying spider bites Identifying spider bi...
1     history of orcas island Orcas Island, located ...
2     tooth abscess A tooth abscess is a painful inf...
3     barrett's esophagus Barrett's esophagus is a p...
4     teddy bears Teddy bears are stuffed toys typic...
5     patron saint of mental illness The patron sain...
6     holes by louis sachar 'Holes' is a novel by Lo...
7     hip roof A hip roof is a type of roof design w...
8     carpenter bee The carpenter bee (Xylocopa spp....
9     the american revolutionary The American Revolu...
10    folk remedies sore throat For a sore throat, s...
11    balding cure Currently, there is no definitive...
12    evidence for evolution The evidence for evolut...
13    tribe formerly living in alabama The Creek Nat...
14    F5 tornado An F5 tornado is the highest rating...
15    symptoms of heart attack The symptoms of a hea...
16    feliz navidad lyrics "Feliz Navidad" is a Chri...
17    benefits of running Running offers numerous ph...
18    marshall county schools Marshall County School...
19    sun tzu Sun Tzu was a Chinese military strateg...
20    halloween activities for middle school For mid...
21    dreams interpretation Dreams interpretation is...
22    wilson's disease Wilson's disease is a rare ge...
23    golf instruction Golf instruction typically in...
24    uss cole The USS Cole (DDG-67) is an Arleigh B...
25    how has african american music influence histo...
26    bewitched cast The main cast of the classic TV...
27    mister rogers Mister Rogers, whose full name w...
28    game theory Game theory is the study of how pe...
29    view my internet history To view your internet...
30    ketogenic diet The ketogenic diet is a low-car...
31    nasa interplanetary missions NASA has conducte...
32    hayrides in pa Pennsylvania offers numerous ha...
33    where to find morel mushrooms Morel mushrooms ...
34    magnesium rich foods Magnesium-rich foods incl...
35    common schizophrenia drugs Common schizophreni...
36    carotid cavernous fistula treatment A carotid ...
```

```
36    carotid cavernous fistula treatment A carotid ...
37    fidel castro Fidel Castro (1926-2016) was a Cu...
38    benefits of yoga Yoga offers numerous physical...
39    norway spruce The Norway spruce (Picea abies) ...
40    sangre de cristo mountains The Sangre de Crist...
41    history of the electronic medical record The h...
42    educational advantages of social networking si...
43    flowering plants Flowering plants, also known ...
44    how to tie a windsor knot To tie a Windsor kno...
45    recycling lead acid batteries Recycling lead a...
46    altitude sickness Altitude sickness, also know...
47    medical care and jehovah's witnesses Jehovah's...
48    pink slime in ground beef Pink slime, also kno...
49    how to find the mean To find the mean, add up ...
Name: query, dtype: object
```

```
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
The tokenizer class you load from this checkpoint is not the same type as the class this
function is called from. It may result in unexpected tokenization.
The tokenizer class you load from this checkpoint is 'ElectraTokenizer'.
The class this function is called from is 'CrossEncoderElectraTokenizerFast'.
```

```
Running Pipeline: BM25_HyDE_CE_Large
calling sliding on df of 5000 rows


Pipeline finished in 0:12:20.494688

                 name        RR   P@10   nDCG@10  nDCG@100
0  BM25_HyDE_CE_Large  0.596172  0.362  0.254356  0.290068
```

Out[14]:

	name	RR	P@10	nDCG@10	nDCG@100
0	**BM25_HyDE_CE_Large**	**0.596172**	**0.362**	**0.254356**	**0.290068**

In [11]:

```
hyde_topics_keywords = generate_hyde_gwdg(
    dataset,
    api_key=GWDG_API_KEY,
    model_name="meta-llama-3.1-70b-instruct",
    system_prompt="You are an expert search engine optimizer. Directly generate 20 highly
relevant keywords and synonyms for the search query.",
    prompt_template="Question: {query}\nPassage:",
    temperature=0.4,
    max_tokens=100
)

# can run the pipeline before, but moved it to the next cell
# since we have limited api calls with the GWDG_API_KEY and we set the topics already
```

```
--- Connecting to GWDG Chat AI (meta-llama-3.1-70b-instruct) ---
--- Generating Hypothetical Documents (Temp=0.4) ---
```

```
Hallucinating: 100%|████████████████████████████████████| 50/50 [02:36<00:00,  3.13s/it
]
```

```
0                              identifying spider bites
1                               history of orcas island
2                                         tooth abscess
3                                  barrett's esophagus
4                                          teddy bears
5                          patron saint of mental illness
6                                   holes by louis sachar
```

```
7                                          hip roof
8                                      carpenter bee
9                             the american revolutionary
10                             folk remedies sore throat
11                                       balding cure
12                                evidence for evolution
13                     tribe formerly living in alabama
14                                          F5 tornado
15                               symptoms of heart attack
16                                   feliz navidad lyrics
17                                benefits of running
18                               marshall county schools
19                                          sun tzu
20               halloween activities for middle school
21                                dreams interpretation
22                                     wilson's disease
23                                     golf instruction
24                                          uss cole
25     how has african american music influence history
26                                      bewitched cast
27                                       mister rogers
28                                         game theory
29                               view my internet history
30                                      ketogenic diet
31                           nasa interplanetary missions
32                                      hayrides in pa
33                           where to find morel mushrooms
34                                magnesium rich foods
35                             common schizophrenia drugs
36                  carotid cavernous fistula treatment
37                                         fidel castro
38                                     benefits of yoga
39                                        norway spruce
40                               sangre de cristo mountains
41              history of the electronic medical record
42    educational advantages of social networking sites
43                                     flowering plants
44                              how to tie a windsor knot
45                           recycling lead acid batteries
46                                    altitude sickness
47                  medical care and jehovah's witnesses
48                               pink slime in ground beef
49                                     how to find the mean
Name: query_original, dtype: object
0     identifying spider bites Here are 20 highly re...
1     history of orcas island Based on the search qu...
2     tooth abscess Here are 20 highly relevant keyw...
3     barrett's esophagus Here are 20 highly relevan...
4     teddy bears Here are 20 highly relevant keywor...
5     patron saint of mental illness Here are 20 hig...
6     holes by louis sachar Based on the search quer...
7     hip roof Here are 20 highly relevant keywords ...
8     carpenter bee Here are 20 highly relevant keyw...
9     the american revolutionary Based on the search...
10    folk remedies sore throat Here are 20 highly r...
11    balding cure Here are 20 highly relevant keywo...
12    evidence for evolution Here are 20 highly rele...
13    tribe formerly living in alabama Based on the ...
14    F5 tornado Here are 20 highly relevant keyword...
15    symptoms of heart attack Based on the search q...
16    feliz navidad lyrics Here are 20 highly releva...
17    benefits of running Here are 20 highly relevan...
18    marshall county schools Here are 20 highly rel...
19    sun tzu Based on the search query "sun tzu", h...
20    halloween activities for middle school Here ar...
21    dreams interpretation Here are 20 highly relev...
22    wilson's disease Here are 20 highly relevant k...
23    golf instruction Here are 20 highly relevant k...
24    uss cole Here are 20 highly relevant keywords ...
25    how has african american music influence histo...
26    bewitched cast Here are 20 highly relevant key...
27    mister rogers Here are 20 highly relevant keyw...
```

```
28    game theory Here are 20 highly relevant keywor...
29    view my internet history Here are 20 highly re...
30    ketogenic diet Here are 20 highly relevant key...
31    nasa interplanetary missions Here are 20 highl...
32    hayrides in pa Here are 20 highly relevant key...
33    where to find morel mushrooms Here are 20 high...
34    magnesium rich foods Here are 20 highly releva...
35    common schizophrenia drugs Here are 20 highly ...
36    carotid cavernous fistula treatment Based on t...
37    fidel castro Here are 20 highly relevant keywo...
38    benefits of yoga Here are 20 highly relevant k...
39    norway spruce Here are 20 highly relevant keyw...
40    sangre de cristo mountains Here are 20 highly ...
41    history of the electronic medical record Based...
42    educational advantages of social networking si...
43    flowering plants Here are 20 highly relevant k...
44    how to tie a windsor knot Here are 20 highly r...
45    recycling lead acid batteries Here are 20 high...
46    altitude sickness Here are 20 highly relevant ...
47    medical care and jehovah's witnesses Based on ...
48    pink slime in ground beef Here are 20 highly r...
49    how to find the mean Here are 20 highly releva...
Name: query, dtype: object

---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[11], line 15
      1 hyde_topics_keywords = generate_hyde_gwdg(
      2     dataset,
      3     api_key=GWDG_API_KEY,
   (...)      8     max_tokens=100
      9 )
     11 # Run the pipeline
     12 run_hyde_experiment(
     13     dataset,
     14     hyde_topics_keywords,
---> 15     SPARSE_PATH,
     16     "webis/monoelectra-large"
     17 )

NameError: name 'SPARSE_PATH' is not defined
```

In [13]:

```
SPARSE_PATH = "./indices/sparse_clueweb"
DENSE_ROOT  = "./indices/dense_clueweb"
BI_MODEL = "webis/bert-bi-encoder"
CE_MODEL = "webis/monoelectra-large"

dense_path = build_indexes(dataset, SPARSE_PATH, DENSE_ROOT, BI_MODEL)

# Run the pipeline
run_hyde_experiment(
    dataset,
    hyde_topics_keywords,
    SPARSE_PATH,
    "webis/monoelectra-large"
)
```

```
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
The tokenizer class you load from this checkpoint is not the same type as the class this
function is called from. It may result in unexpected tokenization.
The tokenizer class you load from this checkpoint is 'ElectraTokenizer'.
The class this function is called from is 'CrossEncoderElectraTokenizerFast'.
```

Running Pipeline: BM25 HyDE CE Large

```
Running Pipeline: BM25_HyDE_CE_Large
calling sliding on df of 5000 rows

Pipeline finished in 0:11:35.727008

                  name        RR   P@10   nDCG@10  nDCG@100
0  BM25_HyDE_CE_Large  0.534964  0.372  0.261261  0.292594
```

Out[13]:

	name	RR	P@10	nDCG@10	nDCG@100
0	**BM25_HyDE_CE_Large**	**0.534964**	**0.372**	**0.261261**	**0.292594**

Energy

In [12]:

```python
import time
import pandas as pd
import pyterrier as pt
from codecarbon import EmissionsTracker

bm25 = pt.terrier.Retriever(index, wmodel="BM25")
ce = CrossEncoder(
    model_name_or_path="webis/monoelectra-large",
    text_field="text",
    batch_size=32,
    use_gpu=True
)
ce._module.to(device)
ce._module.device

# Reconstruct the Full Pipeline
# BM25 (Top 100) -> Fetch Text -> Sliding Window -> CrossEncoder -> Max Passage
pipeline = (
    (bm25 % 100)
    >> pt.text.get_text(dataset, "text")
    >> pt.text.sliding(length=300, stride=150, text_attr='text', prepend_attr=None)
    >> ce
    >> pt.text.max_passage()
)

# Measurement Function
def measure_efficiency(pipeline, topics, run_name="BM25+CE_Large"):
    print(f"--- Measuring Efficiency: {run_name} ---")

    tracker = EmissionsTracker(
        project_name=run_name,
        output_dir="./",
        measure_power_secs=1,
        log_level='error',
        save_to_file=True
    )

    tracker.start()

    start_time = time.time()

    res = pipeline.transform(topics)

    end_time = time.time()
    emissions = tracker.stop()

    total_seconds = end_time - start_time
    latency_ms = (total_seconds / len(topics)) * 1000  # ms per query

    print(f"   Latency:    {latency_ms:.2f} ms/query")
    print(f"   Emissions:  {emissions:.6f} kgCO2")
```

```
    return {"Model": run_name, "Latency": latency_ms, "Emissions": emissions}

# Run Measurement
stats = measure_efficiency(pipeline, dataset.get_topics(variant='text'))

# # Run on just 10 queries to get the numbers instantly
# stats = measure_efficiency(pipeline, dataset.get_topics(variant='text').head(10))
pd.DataFrame([stats])
```

```
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
You are using a model of type mono to instantiate a model of type electra. This is not su
pported for all configurations of models and can yield errors.
The tokenizer class you load from this checkpoint is not the same type as the class this
function is called from. It may result in unexpected tokenization.
The tokenizer class you load from this checkpoint is 'ElectraTokenizer'.
The class this function is called from is 'CrossEncoderElectraTokenizerFast'.
```

```
--- Measuring Efficiency: BM25+CE_Large ---
calling sliding on df of 5000 rows


   Latency:    4724.35 ms/query
   Emissions:  0.013723 kgCO2
```

Out[12]:

	Model	Latency	Emissions
0	**BM25+CE_Large**	**4724.353781**	**0.013723**

In [13]:

```
print(pd.DataFrame([stats]))
```

```
           Model      Latency  Emissions
0  BM25+CE_Large  4724.353781   0.013723
```

Discussion

Use the following marked cells to respond to the following prompts

1. Describe your retrieval model and why you implemented it in the way you did.

(250 words max)

The final model is a Retrieve and Rerank pipeline consisting of BM25 (Sparse Retrieval) followed by a Cross-Encoder (MonoElectra-Large) reranker.

This specific architecture was implemented to navigate the "messy" nature of the ClueWeb dataset:

- **First Stage: BM25 is prioritized over dense retrieval because our experiments revealed a severe domain shift. The `webis/bert-bi-encoder` which is trained on clean questions, failed to handle the noisy web forums of the corpus, often filtering out relevant documents that BM25 successfully caught. Consequently, BM25 proved more robust than either a pure Bi-Encoder or a hybrid (BM25 + Bi-Encoder) approach.**
- **Second Stage: We utilized `webis/monoelectra-large` with a reranking depth of 100. Switching from "base" to "large" yielded a +3.6% boost in RR, suggesting that model capacity was a primary driver of effectiveness for this task.**
- **Context Management: We implemented a sliding window of 300 tokens with a stride of 150. Experiments identified this as the "sweet spot": shorter windows (150) cut off necessary context, while longer ones (450) introduced too much noise from footers and navigation bars.**

Furthermore, Query Expansion via RM3 failed to improve performance, likely by introducing irrelevant terms. Attempting expansion via the `meta-llama-3.1-70b-instruct` LLM also degraded performance. This failure was likely caused by the LLM prefixing the response with conversational filler (e.g., "Here are 20 highly relevant keywords..."), which was then truncated at 150 tokens, effectively injecting garbage data into the search query.

2. Describe how you could improve your model further and any limitations of your retrieval model.

(250 words max)

Limitations:

- Extreme Latency: As measured, the model requires 4,724 ms (4.7s) per query. This makes real-time deployment impossible, as the Cross-Encoder must perform heavy self-attention on 100 documents for every single search.
- Recall Ceiling: The pipeline is strictly bound by the recall of BM25. If a document lacks exact keyword overlap, the Cross-Encoder never sees it.
- Query Expansion Failure: The attempt to fix the recall issue with LLM-based Query Expansion failed (scoring 0.59 vs baseline). This was due to the injection of conversational filler words (as described above) and possibly "hallucination drift," where the LLM's vocabulary did not match the noisy corpus.

Improvements:

- Preprocess Query Expansion: Strip redundant conversational phrases (e.g., "Here are the keywords") from the LLM output to isolate only the effective search terms.
- Stricter System Prompt: Refine the prompt to enforce direct keyword output without conversational filler which will eliminating the need for post-processing.
- Domain Adaptation: Fine-tune the query generator to align hallucinations with the specific ClueWeb domain, reducing the vocabulary mismatch we observed.
- Knowledge Distillation: Distill the Cross-Encoder into a smaller architecture like a Bi-Encoder or Late Interaction model (ColBERT). Training on the teacher's scores would fix the domain shift issue while reducing latency from 4.7s to <100ms via pre-computation.

3. Describe the effectiveness--efficiency trade-offs made by your retrieval model (storage/computation/energy/etc).

see lecture on *Retrieval Efficiency*

(250 words max)

The final model explicitly prioritizes Effectiveness (nDCG@10: 0.375) at a massive cost to Efficiency, placing it deep in the "Red" zone of Green IR evaluation.

- Computation & Latency: Unlike sparse lookups which take milliseconds, the Cross-Encoder performs expensive BERT self attention ($O(N^2)$) on the top 100 documents. The measurements show a staggering latency of 4,724 ms (4.7s) per query, which is approximately 500x slower than a standard BM25 system (<10ms).
- Energy Consumption: By selecting the "Large" architecture, we significantly increased the floating point operations (FLOPs). The EmissionTracker calculated 0.0137 kgCO_2 emissions for our small batch alone.
  - Context: 0.0137 kg (13.7 grams) of CO_2 is roughly equivalent to fully charging a smartphone twice. Generating this much carbon for just a handful of search queries highlights why this architecture is unsustainable for a real world search engine handling millions of users.
- Storage: Conversely, our storage footprint is efficient. By rejecting dense retrieval indices (which require large RAM for vectors), the model relies only on the compact inverted index.

In summary, the model achieved high retrieval quality (0.375 nDCG@10) but created a system that is too slow and energy intensive for practical and large-scale use.